

MAT31-2

Probability & Statistics II

Continuous random variables, limit theorems, and hypothesis testing

From continuous distributions to the Central Limit Theorem and statistical inference.

Albert School — 30 hours

Contents

1	Generalized integrals	2
2	Continuous random variables	3
3	Classic continuous distributions	3
3.1	Uniform	3
3.2	Normal	4
3.3	Exponential	4
3.4	Chi-squared and Student-t	4
4	Markov property and the Bellman equation	5
5	Law of Large Numbers	6
6	Central Limit Theorem	6
7	Monte Carlo methods	6
8	Hypothesis testing	7
9	Confidence intervals	7

1 Generalized integrals

Definition 1 Let f be a function defined on $[a, +\infty)$ and integrable on every $[a, b]$. The **generalized** (improper) integral of f on $[a, +\infty)$ is

$$\int_a^{+\infty} f(x) dx = \lim_{b \rightarrow +\infty} \int_a^b f(x) dx,$$

and it **converges** when this limit exists and is finite. Improper integrals on $(-\infty, b]$ and on an interval where f is unbounded at an endpoint are defined analogously by a limit.

Worked example.

$$\int_1^{+\infty} \frac{1}{x^2} dx = \lim_{b \rightarrow +\infty} \left[-\frac{1}{x} \right]_1^b = \lim_{b \rightarrow +\infty} \left(1 - \frac{1}{b} \right) = 1,$$

so the integral converges. By contrast

$$\int_1^{+\infty} \frac{1}{x} dx = \lim_{b \rightarrow +\infty} [\ln x]_1^b = \lim_{b \rightarrow +\infty} \ln b = +\infty,$$

so it diverges.

Proposition 2 (Reference integrals)

$$\int_1^{+\infty} \frac{1}{x^\alpha} dx \text{ converges} \Leftrightarrow \alpha > 1, \quad \int_0^1 \frac{1}{x^\alpha} dx \text{ converges} \Leftrightarrow \alpha < 1.$$

Also $\int_0^{+\infty} e^{-\lambda x} dx = \frac{1}{\lambda}$ converges for $\lambda > 0$.

This intuitive treatment is sufficient here; the full convergence framework is deferred to the Advanced track (AD S3).

2 Continuous random variables

Definition 3 A random variable X is **continuous** with **probability density function** (PDF) f if $f(x) \geq 0$, $\int_{-\infty}^{+\infty} f(x) dx = 1$, and for every interval,

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

In particular $P(X = x) = 0$ for every single value x .

Definition 4 The **cumulative distribution function** (CDF) of X is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

It is non-decreasing, with limits 0 at $-\infty$ and 1 at $+\infty$, and $F'(x) = f(x)$ at every point where f is continuous.

Definition 5 The **expectation** of a continuous random variable X is

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f(x) dx,$$

when the integral converges absolutely.

Theorem 6 (Transfer theorem) For a function g ,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{+\infty} g(x) f(x) dx,$$

provided the integral converges absolutely.

Definition 7 The **variance** of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2,$$

and its **standard deviation** is $\sqrt{\text{Var}(X)}$. The **k -th moment** of X is $\mathbb{E}[X^k] = \int_{-\infty}^{+\infty} x^k f(x) dx$.

3 Classic continuous distributions

3.1 Uniform

Definition 8 $X \sim \text{Unif}(a, b)$ has constant density on $[a, b]$:

$$f(x) = \frac{1}{b-a} \text{ for } a \leq x \leq b, \quad f(x) = 0 \text{ otherwise.}$$

Its CDF is $F(x) = \frac{x-a}{b-a}$ on $[a, b]$, and

$$\mathbb{E}[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

3.2 Normal

Definition 9 $X \sim \mathcal{N}(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R},$$

with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$. The **standard normal** is $\mathcal{N}(0, 1)$, with density φ and CDF Φ .

Proposition 10 (Standardization) If $X \sim \mathcal{N}(\mu, \sigma^2)$ then $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1)$, and

$$P(X \leq x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

Values of Φ are read from a **standard normal table**.

Proposition 11 (68–95–99.7 rule) For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$P(|X - \mu| \leq \sigma) \approx 0.68, \quad P(|X - \mu| \leq 2\sigma) \approx 0.95, \quad P(|X - \mu| \leq 3\sigma) \approx 0.997.$$

3.3 Exponential

Definition 12 $X \sim \text{Exp}(\lambda)$ ($\lambda > 0$) has density

$$f(x) = \lambda e^{-\lambda x} \text{ for } x \geq 0, \quad f(x) = 0 \text{ otherwise,}$$

with CDF $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$, and

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

Proposition 13 (Memoryless property) For $X \sim \text{Exp}(\lambda)$ and $s, t \geq 0$,

$$P(X > s + t \mid X > s) = P(X > t).$$

The exponential is the only continuous distribution with this property.

3.4 Chi-squared and Student-t

Definition 14 If $Z_1, \dots, Z_k$ are independent $\mathcal{N}(0, 1)$, then

$$\chi_k^2 = Z_1^2 + \dots + Z_k^2$$

follows the **chi-squared** distribution with k degrees of freedom, with $\mathbb{E}[\chi_k^2] = k$ and $\text{Var}(\chi_k^2) = 2k$.

Definition 15 If $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_k^2$ are independent, then

$$T = \frac{Z}{\sqrt{V/k}}$$

follows the **Student- t** distribution with k degrees of freedom. Its density is symmetric and bell-shaped with heavier tails than the normal, approaching $\mathcal{N}(0, 1)$ as $k \rightarrow \infty$.

4 Markov property and the Bellman equation

Definition 16 A sequential model has the **Markov property** if the distribution of the next state depends only on the current state, not on the earlier history. It is specified by a **state space**, a **transition kernel** $P(s' | s, a)$ giving the probability of moving to state s' after taking action a in state s , a **reward** $r(s, a)$, and a **discount factor** $\gamma \in [0, 1)$.

Definition 17 The **value** $V(s)$ of a state is the expected discounted return obtained by acting optimally from s onward. It satisfies the **Bellman optimality equation**

$$V(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right],$$

read as “value today = best reward today + discounted value tomorrow”.

Worked example — Two-state machine: repair or replace. A machine is either **working** (w) or **broken** (b), with discount $\gamma = 0.9$. In state w the only action yields reward 10 and the machine stays in w with probability 0.8, else moves to b . In state b two actions are available:

- **repair**: reward -3 , returns to w with probability 0.6, else stays b ;
- **replace**: reward -8 , returns to w with certainty.

The Bellman equations are

$$\begin{aligned} V(w) &= 10 + 0.9(0.8V(w) + 0.2V(b)), \\ V(b) &= \max \left\{ \underbrace{-3 + 0.9(0.6V(w) + 0.4V(b))}_{\text{repair}}, \underbrace{-8 + 0.9V(w)}_{\text{replace}} \right\}. \end{aligned}$$

Each equation is a one-step accounting identity: immediate reward plus the discounted expected value of the next state.

Solving these equations algorithmically (value iteration, policy iteration, Q-learning) is deferred to the S6 Math-for-ML capstone, and the contraction-mapping proof to the Advanced track.

5 Law of Large Numbers

Theorem 18 (Law of Large Numbers) Let $X_1, X_2, \dots$ be independent and identically distributed with mean $\mu = \mathbb{E}[X_1]$, and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Then

$$\bar{X}_n \rightarrow \mu \quad \text{as } n \rightarrow \infty.$$

The sample mean converges to the true mean as the sample size grows.

Common pitfall The LLN does not say that short-run deviations are “corrected”. After a run of heads, the next toss of a fair coin is still heads with probability $\frac{1}{2}$; the average converges only because later terms dilute early deviations, not because the process compensates for them. Believing otherwise is the **Gambler’s Fallacy**.

6 Central Limit Theorem

Theorem 19 (Central Limit Theorem) Let $X_1, X_2, \dots$ be independent and identically distributed with mean μ and variance $\sigma^2 < \infty$. Then for large n the standardized sum is approximately standard normal:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1),$$

equivalently $\sum_{i=1}^n X_i \approx \mathcal{N}(n\mu, n\sigma^2)$.

Proposition 20 (Normal approximation of the binomial) For $X \sim \text{Bin}(n, p)$ with n large, $X \approx \mathcal{N}(np, np(1-p))$. Because X is integer-valued, apply the **continuity correction**

$$P(X \leq k) \approx \Phi\left(\frac{k + 0.5 - np}{\sqrt{np(1-p)}}\right).$$

To achieve a target margin of error E at confidence level corresponding to z , the required sample size follows from $z\sigma/\sqrt{n} \leq E$, i.e. $n \geq (z\sigma/E)^2$.

7 Monte Carlo methods

A **Monte Carlo** estimate of $\mathbb{E}[g(X)]$ draws independent samples $X_1, \dots, X_n$ and averages:

$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n g(X_i).$$

By the LLN $\hat{\mu}_n \rightarrow \mathbb{E}[g(X)]$, and by the CLT it is approximately normal.

Proposition 21 (Inverse transform method) If F is the CDF of X and $U \sim \text{Unif}(0, 1)$, then $X = F^{-1}(U)$ has CDF F . This produces samples of X from uniform draws.

Worked example — Exponential by inverse transform. For $\text{Exp}(\lambda)$, $F(x) = 1 - e^{-\lambda x}$, so solving $U = F(x)$ gives

$$X = -\frac{1}{\lambda} \ln(1 - U),$$

which is $\text{Exp}(\lambda)$ -distributed.

Definition 22 The **standard error** of the Monte Carlo estimate is $\hat{s}/\sqrt{n}$, where $\hat{s}^2 = \frac{1}{n-1} \sum_{i=1}^n (g(X_i) - \hat{\mu}_n)^2$. An approximate **confidence interval** for $\mathbb{E}[g(X)]$ is

$$\hat{\mu}_n \pm z \frac{\hat{s}}{\sqrt{n}}.$$

8 Hypothesis testing

Definition 23 A **hypothesis test** opposes the **null hypothesis** H_0 to the **alternative** H_1 . A **test statistic** is computed from the data; the test **rejects** H_0 when the statistic falls in the **rejection region**, chosen so that the probability of rejecting a true H_0 equals the **significance level** α .

Definition 24 The **p-value** is the probability, computed assuming H_0 holds, of observing data at least as extreme as that actually observed. One rejects H_0 when the p-value is at most α .

Common pitfall The p-value is “the probability of data this extreme assuming H_0 holds”, *not* “the probability that H_0 is true”. H_0 is not a random event.

Definition 25 A **Type I error** is rejecting a true H_0 (probability α). A **Type II error** is failing to reject a false H_0 (probability β). The **power** of the test is $1 - \beta$, the probability of correctly rejecting a false H_0 .

9 Confidence intervals

Definition 26 Using the CLT, a **confidence interval** of level $1 - \alpha$ for a mean μ from a sample of size n is

$$\bar{X}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}},$$

where $z_{\alpha/2}$ satisfies $\Phi(z_{\alpha/2}) = 1 - \alpha/2$. The quantity $E = z_{\alpha/2} \sigma / \sqrt{n}$ is the **margin of error**.

Common pitfall The level $1 - \alpha$ is a **property of the procedure**: over many samples, a proportion $1 - \alpha$ of the intervals produced contain μ . For one computed interval it is wrong to say “ μ lies in it with probability $1 - \alpha$ ” — once computed, the interval either contains μ or it does not.

Proposition 27 (Duality with hypothesis tests) A value μ_0 lies in the level- $(1 - \alpha)$ confidence interval if and only if the two-sided test of $H_0 : \mu = \mu_0$ is **not** rejected at significance level α .